

0902.0431 — formula report

3170 inline formulas (section omitted; all rendered), 1219 display equations, 14 tables, 11 unrecovered image regions.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00515 (now)	85	—	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00515 (was)	85	■0.002	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	—
0902.0431_E00515 (now)	85	—	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00756 (was)	125	■0.119	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 + \lambda_1 = 1 \\ &\lambda_0 - \lambda_2 = 1 \quad \lambda_0 + \lambda_2 = 1 \\ &\lambda_0 - \lambda_3 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 0 \quad \lambda_1 + \lambda_2 = 0 \\ &\lambda_1 - \lambda_3 = 0 \quad \lambda_1 + \lambda_3 = 0 \\ &\lambda_2 - \lambda_3 = 0 \quad \lambda_2 + \lambda_3 = 0 \end{aligned}$	—
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E08756 (now)	125	—	$\begin{array}{l} \backslash begin{array}{llllllllllll} \backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}=1 \\ \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{1\}}=1 \& \\ 2 \& 2 \& 2 \& 2 \& 2 \& 1 \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{2\}}=1 \& 1 \& \\ 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{2\}}=1 \& 1 \& 2 \\ \& 2 \& 2 \& 1 \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{3\}}=1 \& 1 \& 1 \& 0 \\ \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{3\}}=1 \& 1 \& 1 \& 2 \& \\ 2 \& 1 \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}=0 \& 1 \& 0 \& 0 \& 0 \& \\ 0 \& 0 \& \backslash lambda_{\{1\}}+\backslash lambda_{\{2\}}=0 \& 1 \& 2 \& 2 \& 2 \& 1 \\ \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{3\}}=0 \& 1 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ \& \backslash lambda_{\{1\}}+\backslash lambda_{\{3\}}=0 \& 1 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \\ \backslash lambda_{\{2\}}-\backslash lambda_{\{3\}}=0 \& 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& 0 \& \\ \backslash lambda_{\{2\}}+\backslash lambda_{\{3\}}=0 \& 0 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \& \& \\ \backslash lambda_{\{0\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=1 \\ \& 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \backslash lambda_{\{1\}}+ \\ \frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 1 \& 1 \& 1 \& 1 \& \\ 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \backslash lambda_{\{2\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 1 \& 1 \& 1 \& 1 \& \& \& \\ \& \backslash \backslash \& \& \backslash lambda_{\{3\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 0 \& 1 \& 1 \& 1 \& 1 \& \& \& \& \end{array}$	$\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & & 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 \\ \lambda_0-\lambda_2=1 & 1 & 0 & & 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 \\ \lambda_0-\lambda_3=1 & 1 & 1 & & 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 \\ \lambda_1-\lambda_2=0 & 1 & 0 & & 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 \\ \lambda_1-\lambda_3=0 & 1 & 1 & & 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 \\ \lambda_2-\lambda_3=0 & 0 & 1 & & 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 \\ & & & & \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 & & & & 1 \\ & & & & \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 & & & & 1 \\ & & & & \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 & & & & 1 \\ & & & & \lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 & & & & 1 \end{array}$	—
basis: inferred / ink					
0902.0431_E08922 (was)	153	■ 0.102	$f^{-1}\left(\sum_{i<j<k} x_{\{i\ j\ k\}} \boldsymbol{e}_{\{i\}} \wedge \boldsymbol{e}_{\{j\}} \wedge \boldsymbol{e}_{\{k\}} \right)=\left(\begin{array}{c} \left(\begin{array}{c} \left(\begin{array}{c} h\left(x_{\{156\}}\right) \& h\left(x_{\{164\}}, \bar{x}_{\{256\}} \right) \& h\left(x_{\{145\}}, \bar{x}_{\{356\}} \right) \backslash \backslash h\left(x_{\{256\}}, \bar{x}_{\{164\}} \right) \& h\left(x_{\{264\}} \right) \& h\left(x_{\{245\}}, \bar{x}_{\{364\}} \right) \backslash \backslash h\left(x_{\{356\}}, \bar{x}_{\{145\}} \right) \& h\left(x_{\{364\}}, \bar{x}_{\{245\}} \right) \& h\left(x_{\{345\}} \right) \end{array} \right) \backslash \backslash \left(\begin{array}{c} \left(\begin{array}{c} h\left(x_{\{423\}} \right) \& h\left(x_{\{431\}}, \bar{x}_{\{523\}} \right) \& h\left(x_{\{412\}}, \bar{x}_{\{623\}} \right) \backslash \backslash h\left(x_{\{523\}}, \bar{x}_{\{431\}} \right) \& h\left(x_{\{531\}} \right) \& h\left(x_{\{512\}}, \bar{x}_{\{631\}} \right) \backslash \backslash h\left(x_{\{623\}}, \bar{x}_{\{412\}} \right) \& h\left(x_{\{631\}}, \bar{x}_{\{512\}} \right) \& h\left(x_{\{612\}} \right) \end{array} \right) \right) \backslash \backslash \backslash \left(\begin{array}{c} \left(\begin{array}{c} h\left(x_{\{123\}} \right) \& \backslash \backslash h\left(x_{\{456\}} \right) \end{array} \right) \right) \right) \right) \right).$	(not rendered)	—
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00922 (now)	153	—	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\right)\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{c}\left(\begin{array}{ccc}h\left(x_{156}\right)&h\left(x_{164},\bar{x}_{256}\right)&h\left(x_{145},\bar{x}_{356}\right)\\h\left(x_{256},\bar{x}_{164}\right)&h\left(x_{264}\right)&h\left(x_{245},\bar{x}_{364}\right)\\h\left(x_{356},\bar{x}_{145}\right)&h\left(x_{364},\bar{x}_{245}\right)&h\left(x_{345}\right)\end{array}\right)\\\left(\begin{array}{ccc}h\left(x_{423}\right)&h\left(x_{431},\bar{x}_{523}\right)&h\left(x_{412},\bar{x}_{623}\right)\\h\left(x_{523},\bar{x}_{431}\right)&h\left(x_{531}\right)&h\left(x_{512},\bar{x}_{631}\right)\\h\left(x_{623},\bar{x}_{412}\right)&h\left(x_{631},\bar{x}_{512}\right)&h\left(x_{612}\right)\end{array}\right)\end{array}\right)\\\left(\begin{array}{cc}h\left(x_{123}\right)&\\&h\left(x_{456}\right)\end{array}\right)\end{array}\right),$	<i>(not rendered)</i>	—
basis: inferred / ink					
0902.0431_E01032 (was)	173	■ 0.022	$\begin{aligned}&\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1\quad 2\quad 3\quad 3\quad 2\quad 2\end{aligned}\quad 1\quad \square=\frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1\quad 2\quad 3\quad 3\quad 2\quad 2\quad 1\quad 1\quad 1\quad 1.$	<i>(not rendered)</i>	—
0902.0431_E01032 (now)	173	—	$\begin{aligned}\kappa\pi H&=\tfrac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)H_0+\tfrac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1\\&\quad +\tfrac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\tfrac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3,\end{aligned}$		—
basis: inferred / ink					
0902.0431_E01059 (was)	180	■ 0.093	$\begin{aligned}&\&=\left\{\alpha\in E_8\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\}\\&=\left\{\alpha\in E_7^C\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\}\text{ (Proposition 5.7.1)}\\&=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)}\\&=E_7\text{ (Lemma 4.3.3.(4)).}\end{aligned}$		—
0902.0431_E01059 (now)	180	—	$\begin{aligned}\kappa\pi H&=\tfrac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)H_0+\tfrac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1\\&\quad +\tfrac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\tfrac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3,\end{aligned}$		—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01117 (was)	189	0.108	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left(\left(X_1, Y_1 \right) + \left(X_2, Y_2 \right) + \left(X_3, Y_3 \right) \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in \left(\mathfrak{J}^C \right)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee \boldsymbol{Y} & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	—	
0902.0431_E01117 (now)	189		$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left(\left(X_1, Y_1 \right) + \left(X_2, Y_2 \right) + \left(X_3, Y_3 \right) \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \begin{pmatrix} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{pmatrix} \in \left(\mathfrak{J}^C \right)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	—	

basis: inferred / ink

Conf.

MathPix confidence:

≥0.9

0.5-0.9

<0.5

— no value

Residual

render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1187 (was)	200	■ 0.183	$\begin{aligned} \&=(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \\ &\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\left(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})\right) \&+ \\ &\left(\boldsymbol{a}_2, \boldsymbol{w}\right) \\ &\left(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})\right) - \\ &\frac{3}{5}*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \&= \\ &\frac{2}{5}*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - \\ &\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&+ \left(\boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{w}\right) \left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \right. \\ &\left. \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&((\\ &\boldsymbol{b} \times \text{⊞} \text{⊞} - \text{⊞} \text{⊞} - \text{zh20} \\ &\&=(\boldsymbol{a} \wedge \boldsymbol{x})) \\ &\boldsymbol{v}, \boldsymbol{w}) \&=(\boldsymbol{v}, \\ &\boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{v})+\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\right) \left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \right. \\ &\left. \boldsymbol{w} \wedge \boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&- \left(\boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{w}\right) \left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \right. \\ &\left. \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) - \frac{3}{5}*(\\ &\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\\ &\boldsymbol{v}, \boldsymbol{w}) . \end{aligned}$	(not rendered)	—
0902.0431_EQ1187 (now)	200	—	$\begin{aligned} H \&= \nu \Big(\sum_{k=0}^3 \lambda_k H_k \\ &\&= \tfrac{1}{2} \big(\lambda_0 (-H_0 + H_1 - H_2 + H_3) \\ &\quad + \lambda_1 (-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) \\ &\quad + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \big) \\ &\&= \tfrac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 \\ &\quad + \tfrac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ &\quad + \tfrac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 \\ &\quad + \tfrac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3 \end{aligned}$	$\begin{aligned} H &= \nu \Big(\sum_{k=0}^3 \lambda_k H_k \Big) \\ &= \tfrac{1}{2} (\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3)) \\ &= \tfrac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 + \tfrac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ &\quad + \tfrac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \tfrac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3, \end{aligned}$	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved (4)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00037	011	—	$\begin{array}{l} \begin{array}{ll} 1 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ \nu \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu\pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{array}$	(not rendered)	
0902.0431_E00696	115	—	$\begin{array}{l} \begin{array}{cc} \Phi_{-3} = \left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \end{array} \right), \quad \Phi_3 = \left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \end{array} \right), \\ \Phi_{-2} = \left(\begin{array}{cccc} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array} \right), \quad \Phi_2 = \left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right), \\ \Phi_{-1} = \left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array} \right), \quad \Phi_1 = \left(\begin{array}{cccc} 0 & 0 & 0 & A \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right), \\ \Phi_0 = \left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array} \right). \end{array}$	(not rendered)	
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00802	134	—	$\left(\begin{array}{c} \left(\begin{array}{ccc} \cos a_1 a_2\frac{\sin a_2 }{ a_2 }a_3\frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1\frac{\sin a_1 }{ a_1 }\cos a_2 a_3\frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1\frac{\sin a_1 }{ a_1 }a_2\frac{\sin a_2 }{ a_2 }\cos a_3 \end{array}\right) \\ \left(\begin{array}{ccc} a_1\frac{\sin a_1 }{ a_1 }\cos a_2 \cos a_3 & 0 & 0 \\ 0 & \cos a_1 a_2\frac{\sin a_2 }{ a_2 }\cos a_3 & 0 \\ 0 & 0 & \cos a_1 \cos a_2 a_3\frac{\sin a_3 }{ a_3 } \end{array}\right) \\ \cos a_1 \cos a_2 \cos a_3 \\ a_1\frac{\sin a_1 }{ a_1 }a_2\frac{\sin a_2 }{ a_2 }a_3\frac{\sin a_3 }{ a_3 } \end{array}\right)$	(not rendered)	$\left(\begin{array}{c} \left(\begin{array}{ccc} \cos a_1 a_2\frac{\sin a_2 }{ a_2 }a_3\frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1\frac{\sin a_1 }{ a_1 }\cos a_2 a_3\frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1\frac{\sin a_1 }{ a_1 }a_2\frac{\sin a_2 }{ a_2 }\cos a_3 \end{array}\right) \\ \left(\begin{array}{ccc} a_1\frac{\sin a_1 }{ a_1 }\cos a_2 \cos a_3 & 0 & 0 \\ 0 & \cos a_1 a_2\frac{\sin a_2 }{ a_2 }\cos a_3 & 0 \\ 0 & 0 & \cos a_1 \cos a_2 a_3\frac{\sin a_3 }{ a_3 } \end{array}\right) \\ \cos a_1 \cos a_2 \cos a_3 \\ a_1\frac{\sin a_1 }{ a_1 }a_2\frac{\sin a_2 }{ a_2 }a_3\frac{\sin a_3 }{ a_3 } \end{array}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00939	156	—	$\left. \begin{array}{l} \xrightarrow{\stackrel{f^{-1}}{\longrightarrow}} \\ \left(\begin{array}{ccc} h\left(b_{11} - c_{11} - \frac{\nu}{3}\right) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right) \\ = \begin{pmatrix} \phi_C(B, C)E_1 + \frac{i\nu e_1}{3}E_1 \\ 2h(L) \times E_1 \\ 0 \\ -(\tau h(L), E_1) \end{pmatrix} = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} E_1 \\ 0 \\ 0 \\ 0 \end{pmatrix} .$	(not rendered)	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on — 1 of 233 shown

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ0093 ● C +76	019	■ 0.382	$\begin{aligned} H_{\{1} \ H_{\{1} \ e_{\{2} \} \ \& \ =H_{\{1} \ e_{\{3} \}=-e_{\{2} \}, \\ \quad \end{aligned}$	$H_1H_1e_2 = H_1e_3 = -e_2,$	$\begin{aligned} H_1H_1e_2 &= H_1e_3 = -e_2, & H_1H_1e_3 &= -H_1e_2 = -e_3, \\ H_1H_1e_4 &= -H_1e_5 = -e_4, & H_1H_1e_5 &= H_1e_4 = -e, \\ H_1H_1e_i &= 0 \text{ otherwise.} \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

The remaining 232 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 30 are component (C), 202 weak (W). By MathPix confidence: 142 at ≥ 0.9 , 49 in 0.6–0.9, 41 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥ 0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.